Ramiro Plüss

Physicist | PhD Candidate in Engineering | Computational Neuroscience, Connectomics & Bio-inspired Robotics

Buenos Aires, Argentina rpluss@itba.edu.ar Website Google Scholar LinkedIn GitHub

Research profile

My research focuses on the bidirectional relationship between network structure, neural dynamics, and behavior, combining computational neuroscience, network models, and bio-inspired robotics to study proprioception and embodied control. I build bio-inspired robotic testbeds to study embodied motor control with computational neuroscience models.

Education

PhD in Engineering (in progress) Jul 2023 – Present

Buenos Aires Institute of Technology (ITBA), Buenos Aires, Argentina

Thesis: "Computational and Robotic Models for Proprioception". Supervisor: Dr. Pablo Martín Gleiser.

Licentiate in Physics (5-year program) Feb 2016 – Mar 2023

National University of Rosario (UNR–FCEIA), Rosario, Argentina

Thesis: "Dynamics and Structure in Adaptive Neural Networks". Supervisor: Dr. Pablo Martín Gleiser.

Thesis repository (in Spanish): UNR institutional repository.

Publications and Preprints

Ramiro Plüss, Hernán Villota, Patricio Orio, "Hemispheric-Specific Coupling Improves Modeling of Functional Connectivity Using Wilson–Cowan Dynamics," in *Computational Neuroscience (LAWCN 2025)*, A. Talevi and V. Rosa Cota (eds.), CCIS, vol. 2734, pp. 109–121, Springer, Cham (2026). DOI: 10.1007/978-3-032-14664-9_10. Preprint: arXiv:2506.22951 (2025).

Ramiro Plüss, Pablo Martín Gleiser, "The Role of Connection Density in an Adaptive Network with Chaotic Units," *Nonlinear Science*, vol. 9, article 100215 (2026). DOI: 10.1016/j.nls.2026.100215. Preprint: arXiv:2505.11437 (2025).

Research Projects

Comparative analysis of larval and adult Drosophila motor circuits Sep 2026 – Present

ITBA (Argentina), BSMS–University of Sussex (UK). Comparative connectomic analysis of premotor and motor circuits, investigating their structural organization and functional principles underlying locomotion across developmental stages.

Whole-brain WC/DMF modeling: hemispheric coupling for multi-scale SC–FC Dec 2025 – Present

ITBA (Argentina), IUE (Colombia), UV/CINV (Chile). Modeling structural–functional connectivity relationships using Wilson–Cowan and Dynamic Mean Field models, focusing on hemispheric-specific coupling and multi-scale connectome analysis.

Connectome-informed bio-inspired robotics Mar 2025 – Present

Development of a hexapod robotic platform to investigate connectome-informed neural control, locomotor pattern generation, and embodied sensorimotor interactions.

UNR–CRH linkage project (IFIR, UNR–CONICET) Aug 2021 – Oct 2022

Experimental characterization of gamma-irradiated red blood cells using laser diffractometry to investigate membrane viscoelastic properties.

Research Visits

Visiting Researcher – Brighton and Sussex Medical School, University of Sussex, UK Sep 2026

Host: Dr. Jimena Berni. Comparative analysis of larval and adult *Drosophila* motor circuits, focusing on how locomotor circuit organization changes across development.

Fellowships, Scholarships & Travel Awards

CONICET Ph.D. Scholarship at ITBA (Supervisor: Dr. Pablo Martín Gleiser) — Apr 2023 – Present

The Company of Biologists Travelling Fellowship, University of Sussex, UK — Sep 2026

Open Software Summer School, University College London, UK — Aug 2026

Konstanz School of Collective Behaviour, University of Konstanz, Germany — Jul – Aug 2026

School on the Origins of Life, Behavior and Cognition, ICTP-SAIFR, Brazil — Mar 2025

School on Synchronization: From Collective Motion to Brain Dynamics, ICTP-SAIFR, Brazil — Feb 2025

Latin American Summer School of Computational Neuroscience, UV, Chile — Jan 2025

IRCN/Chen Institute Joint Course on Neuro-inspired Computation, University of Tokyo, Japan — Jul 2024

Latin American School of Computational Neuroscience, USP, Brazil — Jan 2024

Manuel Belgrano Scholarship — Apr 2022 – Feb 2023

Invited Talks & Research Seminars

Research Seminar, "From physics to neural circuits and bioinspired robotics." Work in Progress (WiP) Seminar Series, Brighton and Sussex Medical School (BSMS), University of Sussex, UK, Sep 2026.

Young Scholar Speaker, "Hemispheric-Specific Coupling Improves Modeling of Functional Connectivity Using Wilson–Cowan Dynamics." V Latin American Workshop on Computational Neuroscience (LAWCN'25), Universidad Nacional de La Plata (UNLP), La Plata, Argentina, Nov 2025.

Invited Speaker, "Why Complex Networks?" International Congress INNOVA GISS 2025: Management of Social and Sustainable Research, Institución Universitaria de Envigado (IUE), Envigado, Colombia, Aug 2025.

Technical Skills & Languages

Programming: Python, Fortran, C/C++ (Arduino).

Scientific Computing: NumPy, SciPy, Numba, NetworkX, Matplotlib.

Connectomics & Visualization: CATMAID, neuPrint, napari.

Robotics & Tools: Arduino, Raspberry Pi, electronics prototyping, Git, LaTeX.

Languages: Spanish (native), English (B2, professional working proficiency).

Teaching Experience

Teaching Assistant, ITBA — Bioinspired Robotics course. — Sep 2024 – Present

Private Tutor — Physics, mathematics, and chemistry for high-school and university students. — 2020 – 2024

Selected Advanced Courses

Spring School on Deep Learning, UBA, Buenos Aires, Argentina — Oct 2025

Artificial Intelligence Systems, ITBA, Buenos Aires, Argentina — Aug – Nov 2024

Complex Networks, ITBA, Buenos Aires, Argentina — Aug – Nov 2024

Modeling and Analysis of Complex Systems, UBA, Buenos Aires, Argentina — Jul 2024

Bioinspired Robotics, ITBA, Buenos Aires, Argentina — Aug – Dec 2023

Poster Presentations in Conferences and International Schools

Ramiro Plüss, "From Neural Structure to Dynamics and Embodied Motor Control." OSSS 2026, UCL, London, UK (Aug 2026); Workshop on Interdisciplinary and Complex Systems, Buenos Aires, Argentina (Oct 2026).

Ramiro Plüss, Hernán Villota, Patricio Orio, "Hemispheric-Specific Coupling Improves Modeling of Functional Connectivity Using Wilson–Cowan Dynamics." LAWCN 2025, La Plata, Argentina (Nov 2025).

Ramiro Plüss, Hernán Villota, Tomás Bosch, Patricio Orio, "Modeling Functional Connectivity Alterations in Schizophrenia Patients from the Structural Connectome." XIV National Congress and XV International Seminar of Neurosciences, Bogotá, Colombia (Apr 2025).

Ramiro Plüss, Pablo Martín Gleiser, "Phase Transition in an Adaptive Network with Chaotic Units." ICTP-SAIFR, São Paulo, Brazil (Feb–Mar 2025); University of Tokyo, Japan (Jul 2024); RAFA 108, Bahía Blanca (Sep 2023); RAFA 107, Bariloche (Sep 2022); Science, Technology and Innovation Conference, Rosario (Oct 2022).

Florencia Gómez Fava, **Ramiro Plüss**, et al., "Preliminary Study of the Viscoelastic Properties of Red Blood Cells Irradiated with Gamma Radiation Using a Rheometer." RAFA 107, Bariloche, Argentina (Sep 2022); Science, Technology and Innovation Conference, Rosario, Argentina (Oct 2022).

Scientific Outreach & Volunteering

Bioinspired Robotics Outreach, ITBA – Demonstrations and presentations at ITBA Future Day, MiniLabs, Goethe Schule visits, and donor events. 2025 – 2026

OATec – STEM outreach and event support at the Argentine Technology Olympiad. Oct 2023

Scientific Communication Project, CAB–CNEA – Scientific photography. Sep 2022

Industry Experience

Capital Market Operator at Rosental Inversiones, Rosario, Argentina Oct – Dec 2022

For extended details and official links to programs, talks, posters and volunteering activities, please visit: plussramiro.github.io

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